Course Plan: Bioinformatics Programming with Python

Instructor: Mirna Mokhtar

Department: Azm Center

Duration: 4 Days

Dates: [21-04-2026] [23-04-2026] [28-04-2026] [30-04-2026]

Time : 10:00 -12:00

Student List

No.	Student Name	Specialization	Level
1	Aya El Assaad	Biotechnologie	PHD
2	Hasna Fawaz Abbas	Microbiologie	PHD
3	Aysha Sayyed	Microbiologie	PHD
4	Fatima Fayad	Microbiologie	PHD
5	Rima Jaber	Molecules Bioactives	PHD
6	Helene Chahine	Microbiologie	PHD
7	Salam Zohbie	Biotechnologie	Master
8	Assma Allouch	Microbiologie	Master
9	Norhan Massri	Biotechnologie	Master
10	Rim El Mir	Microbiologie	Master
11	Joudy Hantour	Biotechnologie	Master
12	Nazih Lazkani	Microbiologie	Master

Training Schedule

Day	Topics	Detailed Content	Learning Objectives	Tools / Activities
Day 1	**Basic Python for Bioinformatics**	Introduction to Python; variables and data types; lists and dictionaries; control flow (loops, conditions); functions; basic data handling	Build foundational programming skills and prepare for bioinformatics applications	Python
Day 2	**Gene Expression Analysis**	Simulation of gene expression data; differential expression (Log_2 fold change); gene filtering; GC content analysis	Analyze and interpret gene expression data and identify significant genes	Python, Pandas, NumPy
Day 3	**Sequence Analysis Pipeline**	NCBI search; FASTA retrieval; parsing; metrics computation; N50 calculation; protein translation; motif analysis	Perform sequence-level analysis of selected genes	NCBI, Biopython
Day 4	**Alignment & Phylogenetics Integration**	Pairwise and multiple sequence alignment; phylogenetic tree construction; data export; biological interpretation; limitations and validation	Integrate sequence and expression results into a complete biological insight	Alignment tools, phylogenetic software

Course Objectives

- Develop advanced skills in Python for bioinformatics
- Perform gene expression analysis and identify key biological signals
- Conduct sequence analysis and functional interpretation
- Build integrated pipelines combining expression and sequence data
- Interpret results in a PhD-level research context

Program Outcome

Participants will build a complete bioinformatics pipeline starting from gene expression analysis to sequence interpretation and phylogenetic validation, producing scientifically meaningful insights suitable for advanced research.
